

Cryptography Assignment - 5

Rail Fence

1. Encrypt the message "Enjoy your journey" using Rail Fence with depth 2 and 3.

Ans: Depth 2 —

E J Y O R O R E
N O Y U J U N Y

Cipher Text: EJYORORE NOYUJUNY

Depth 3 —

E Y R R X
N O Y U J U N Y
J O O E

Cipher Text: EYRRX NOYUJUNY JOOE

2. Encrypt the message as per your choice using Rail Fence with depth 2 and 3.

Plain Text: LOVELY DANCE

Ans: Depth 2 —

L V L D N E
O E Y A C X

Cipher Text: LVLDNE OEYACX

Depth 3 —

L L N
O E Y A C
V D E

Cipher Text: LLN OEYAC VDE

3. Decrypt the cipher "EOCSNYUWLEJYREASNS", encrypted using Rail Fence with depth 4.

Ans: Depth: 4 Length: 19

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	E				O							C						S	
2		N			Y		U				W		L					E	
3			J		Y			R		E				A		S			
4				O					N						S				

Plain Text: ENJOY YOUR NEW CLASSES

4. Decrypt the cipher "BÖÖIQUYUBOSNUUTYRKAS", encrypted using Rail Fence with depth 3.

Ans: Depth: 3 Length: 20

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	B			O					O				I				Q			
2		U		Y		U		B		O		S		N		U		U		T
3			Y				R				K				A				S	

Plain Text: BUY YOUR BOOKS IN AUGUST.

5. Decrypt the cipher "SESNERTISOOCMIO" encrypted using Rail Fence with depth 3. [Corrected Question]

Ans: Depth: 3 Length: 15

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	S			E					S				N		
2		E		R		T		I		S		O		O	
3			C				M				I				O

Plain Text: SECRET MISSION OO.

1. Encrypt the following message "We are discovered" using the key "53124".

5 3 1 2 4
W E A R E
D I S C O
V E R E D

DES

1. Using the Initial Permutation Table (Given), determine the output of the initial permutation box when the input is given in hexadecimal as: 0x0002 0000 0000 0001.

Binary: 0000 0000 0000 0010

$\dots \dots \dots$

$0000 \quad 0000 \quad 0000 \quad 0001$

I/p to box
(Plaintext)

0000 0000 0000 00010 - - - - - 0000 0000 0000

0001 $\uparrow$ 15th Bit.

↑ 64th bit

Input Bit-64 = 1, By mapping out in the ~~the~~ IPT,

Input Bit. 15 = 1, Output Position 25 = 1

Output Position 63 = 1.

64-bit output:

$$\begin{array}{cccccccccccc} \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} & \frac{1000}{8} & \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} & \frac{0000}{0} \\ 0000 & 0000 & 0000 & 0000 & 0000 & 0 & 8 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 2 & & & & & & \end{array}$$

Output in hexadecimal:

0X0000 0080 0000 0002

2. Given the 4-bit binary PlainText(M) 1101 and the Key(K) 1011, calculate the resulting CipherText(C) using a bitwise Exclusive-OR (XOR) operation.

Show your bit-by-bit work.

- Plaintext(M) : 1101 (binary)
- Key(K) : 1011 (binary).

Ans:

1 1 0 1

1 0 1 1

CipherText, $C = 0110$ //

3. The input to S-box 1 is 000000. What is the output?

Ans:

00000

1st 6th
↑ ↑

1st & 6th bit → row [00] = 0

Mid. 4 bits → 0000 = 0.

So, the 0th Row, 0th Column in the S-box is 14.

Binary Convⁿ: of 14 is 1110

Output = 1110 //

* Product Cipher

→ A complex encryption method that combines two or more simpler ciphers (usually substitution and transposition) in sequence to create a much more secure ciphertext than any single cipher could achieve.

Example: Plain Text: HELLO

Step 1: (Substitution) — Using Caesar Cipher,

$H \rightarrow K, E \rightarrow H, L \rightarrow O, L \rightarrow O, O \rightarrow R.$

Intermediate Cipher Text: KHOOR.

Step 2 (Transposition) — Using Rail Fence,

KHOOR
 $\begin{array}{c} K \quad O \quad R \\ \diagdown \quad \diagup \\ H \quad O \\ \diagup \quad \diagdown \end{array}$

Cipher Text: ~~KOKH~~ KORHO

→ In modern cryptography, doing this process once isn't enough. Instead, block ciphers perform these steps over and over again in cycles called 'Rounds'

Ex: DES & AES.

→ Drawbacks:

- High Computational Overhead
- Slower Execution Speeds.
- Vulnerability to Implementation Attacks.
- Complex Key Management.

* ADFGVX Cipher

It is a World War I German field cipher introduced in 1918 that uses a 6x6 Polybius square for substitution followed by columnar transposition with the letters A, D, F, G, V & X.

Example - PlainText: ATTACK 1

Step-1: Polybius Square : — (Ignore Spaces).

	A	D	F	G	V	X
A	a	b	c	d	e	f
D	g	h	i	j	k	l
F	m	n	o	p	q	r
G	s	t	u	v	w	x
V	y	z	0	1	2	3
X	4	5	6	7	8	9

Step-2: Convert PlainText to Coordinates —

ATTACK 1

A → Row A Col A → AA

T → GD

T → GD

A → AA

C → AF

K → DV

1 → VG ⇒ Int. Cipher Text : AAGDGDAAAFDV VG.

Step-3: Columnar Transposition — Keyword-COW

* (If DOG, arrange DGO).

C	O	W
A	A	G
D	G	D
A	A	A
F	D	V
V	G	

Arrange acc. col and Final Cipher Text is :

AAGDGDAAAFDV VG.